

1. mod 4 $\rightarrow [3 \text{ } \textcircled{1} \text{ } 3 \text{ } 3]$; DFT = $[12 \text{ } \textcircled{2} \text{ } 0 \text{ } 0 \text{ } 0]$, IDFT = $[3 \text{ } 3 \text{ } 3 \text{ } 3] \textcircled{2}$
 $\textcircled{1} \textcircled{2} = 3 \textcircled{1}$

• $(??) = 4$ (circ conv. same length o/p) $\textcircled{2}$

• $X(k) = \sum_{n=0}^3 x(n) e^{-j(\pi/2)kn} \Rightarrow x(0) = 4, x(2) = 4, x(1) = -2, x(3) = 6$
 $\Rightarrow (???) = x(0) + x(2) = 4 + 4 = 8 \textcircled{4}$

No Marks for directly writing the values.

2. step 5 $\rightarrow$ as $q^2 \Rightarrow sq$. not $\Rightarrow \pm q \Rightarrow (\pm 0.5) = (\pm 0.5) \textcircled{3}$

3. (a) May: differentiable geometric points. $\textcircled{3}$

(b) CT sine wave whose freq. $\rightarrow f_s/2 \textcircled{3}$

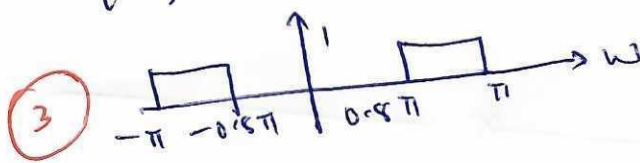
(c) IFFT $\rightarrow 0.25j, 0.75, -0.25j, 0.25 \textcircled{1}$

(d) $[6-a, 1, 0, 1, 6, 1, 0, 1, a]$ & $[6-a, 1-b, 0, 1, 6, 1, 0, 1, a, b]$ with any values of a & b $\textcircled{2}$

(e) NOT necessarily. $\textcircled{3}$

detailed explanation Required

4. $H_1(e^{j\omega}) = \begin{cases} 0, & |\omega| < 0.8\pi \\ 1, & 0.8\pi \leq |\omega| < \pi \end{cases} \textcircled{3}$



HPF $\textcircled{2}$

5. NOT recoverable $\rightarrow$ Time domain aliasing } detailed explanation as in qsn of problem set $\textcircled{8}$
 $x_1[n] = [2 \text{ } 2 \text{ } 1 \text{ } 1 \text{ } 1 \text{ } 1] \textcircled{8}$

6. Performs circular convolution by means of linear convolution explanation of each step for equivalence. $\textcircled{4}$

$z[n] = z[n] + \begin{cases} z[n+N] & ; n < N-1 \\ 0 & ; n = N-1 \end{cases} \textcircled{4}$

7. $g_2[n] = g_1[-(n-3)] + g_1[n-4] \Rightarrow e^{-j3\omega} a_1(e^{-j\omega}) + e^{-j4\omega} a_1(e^{j\omega}) \textcircled{4}$

8. Yes works as claimed (step by step explanation needed) $\textcircled{4}$

Marking scheme

$$Q1 = 12$$

$$Q2 = 3$$

$$Q3 = \begin{matrix} \textcircled{a} & 3 \\ \textcircled{b} & 3 \\ \textcircled{c} & 4 \\ \textcircled{d} & 4 \\ \textcircled{e} & 3 \end{matrix}$$

$$Q4 = 8$$

$$Q5 = 8$$

$$Q6 = 4$$

$$Q7 = 4$$

$$Q8 = 4$$

60 Marks

$$(?) = 3$$

$$(?) = 4$$

$$(?) = 8$$

X